# MKUFT Mathematical Appendix

Professional reading edition of the evolving public MKUFT canon

**Mark Charles McLaughlin**

ORCID: 0009-0005-7736-1511

| | |
|---|---|
| **Scientific affiliation** | Independent Researcher |
| **Object status** | Research module reading edition — not a standalone Zenodo publication |
| **Public source** | Open canonical public source |
| **Source commit** | `26e06b6fea52e6b052dbdd540e80d1d926b96b07` |
| **MKUFT backbone DOI** | 10.5281/zenodo.17780566 |

**Status boundary.** This PDF is a reader-facing rendering of an evolving GitHub research module. It is not a frozen standalone paper and does not receive a separate DOI. The cited Git commit fixes the exact source identity of this edition.

# A1. Substrate as measure space

The substrate scaffold is represented as a measure space,

$$S = (\Omega, \Sigma, \mu),$$

where $\Omega$ is a set of possible configurations, $\Sigma$ a sigma-algebra of measurable subsets, and $\mu$ a baseline measure, weight, or propensity.

The scaffold becomes a probability space only when

$$\mu(\Omega) = 1.$$

The substrate is not directly observed. This is a mathematical representation of structured possibility rather than evidence that a separate material medium has been detected.

# A2. Information structures

Because $I$ names the information layer, the mathematical function space is denoted by

$$\mathcal{I} = L^2(\Omega, \mu).$$

An element $i \in \mathcal{I}$ represents one candidate information structure over the substrate. Such structures may encode patterns, constraints, correlations, and proto-physical forms.

For event $E$, define

$$\mathcal{I}_E \subseteq \mathcal{I}$$

as the set of information structures compatible with $E$ under the active model.

# A3. Physical dynamics

The physical layer supplies an accepted dynamical contribution,

$$P_{\text{phys}}(E),$$

interpreted as a probability, probability density, or other prediction from the relevant established physical model. The mathematical type must be declared for the active event space.

# A4. Observer system and coherence functional

Let the observer state be $\rho_O$. A bounded observer-state coherence functional may be written

$$\kappa : \text{States}(\mathcal{H}_O) \to [0, 1].$$

High $\kappa$ denotes lower measured internal noise or contradiction according to a predefined proxy. The functional is not fixed to one measurement method and requires operational definition in each experiment.

# A5. Integral realisation scaffold

For event $E$, define the unnormalised working weight

$$\widetilde{W}(E) = \int_{i \in \mathcal{I}_E} D_{\text{phys}}(E \mid i) W_{SI}(i \mid S, E) C_O(O \mid i, E) \, d\nu(i),$$

where $D_{\text{phys}}(E \mid i)$ is the accepted physical contribution conditioned on information structure $i$, $W_{SI}(i \mid S, E)$ a candidate substrate-to-information weighting, $C_O(O \mid i, E)$ a bounded observer-condition term, and $\nu$ a measure over compatible information structures.

Where the expression is used as a probability weight, the integrand must be non-negative and the normalisation finite.

For a discrete event space $\mathcal{E}$,

$$P_{\text{realized}}(E) = \frac{\widetilde{W}(E)}{Z},$$

with

$$Z = \sum_{E' \in \mathcal{E}} \widetilde{W}(E').$$

For a continuous event space, the sum is replaced by an integral over a declared event-space measure.

This is a theoretical scaffold. It becomes empirical only when its terms are independently operationalised.

## A6. Reduction to standard physics

If the observer term is constant,

$$C_O(O \mid i, E) = C_0,$$

that constant cancels under normalisation. Standard-physics recovery additionally requires the substrate-to-information weighting and compatible-structure integral to reproduce, or reduce to, the accepted physical distribution:

$$P_{\text{realized}}(E) \approx P_{\text{phys}}(E)$$

within a declared regime, comparison norm, and tolerance.

Failure to recover the ordinary limit is a failure of the physics-facing model.

## A7. Linear-response approximation

A small observer-linked term may be written

$$C_O(O \mid i, E) = C_0[1 + \varepsilon g_O(i, E)],$$

with

$$C_0 > 0, \qquad |\varepsilon g_O(i, E)| < 1$$

throughout the tested domain so that the weight remains positive.

After consistent expansion of numerator and normalisation, a first-order form may be written

$$P_{\text{realized}}(E) \approx P_{\text{phys}}(E) + \varepsilon \Delta_O(E).$$

The sign, scale, and form of $\Delta_O$ must be specified before a confirmatory test, and neglected higher-order terms must remain bounded.

## A8. Coherence-linked modulation

One candidate form is

$$C_O(O \mid i, E) = C_0[1 + \varepsilon \kappa(\rho_O) h(i, E)],$$

where $\kappa$ measures the selected observer-state proxy and $h$ specifies a predefined interaction with event structure. Positivity must hold throughout the tested domain.

Low, irrelevant, or experimentally unsupported $\kappa$ should return the model toward the standard-physics limit.

## A9. Environmental damping

Let $F_{\text{env}}$ denote measured environmental parameters such as electromagnetic noise, geomagnetic indices, geometry, shielding, vibration, temperature, or preregistered material composition.

A simplified form is

$$\kappa_{\text{eff}} = \kappa(\rho_O)\,\eta(F_{\text{env}}),$$

with

$$\eta(F_{\text{env}}) \in [0,1].$$

$\eta$ must be specified or estimated from measurements rather than assigned retrospectively to rescue a null result.

## A10. Group coherence

For observers $\{O_1, \ldots, O_N\}$,

$$\kappa_{\text{group}} = \Phi(\kappa_1, \ldots, \kappa_N, C_{\text{corr}}),$$

where $C_{\text{corr}}$ represents measured alignment or correlation and the range of $\Phi$ must be declared.

A simple bounded candidate approximation is

$$\kappa_{\text{group}} = A_{\text{align}} \frac{1}{N} \sum_{i=1}^{N} \kappa_i,$$

with

$$A_{\text{align}} \in [0,1], \qquad \kappa_i \in [0,1].$$

Group coherence is a coupled O/I/P condition rather than a fifth ontological layer.

## A11. Experimental connections

Candidate application classes include REG/RNG condition comparisons, strictly blinded remote-information protocols, preregistered co-occurrence studies, environmental modulation tests, and group-alignment tests.

These are proposed tests rather than established confirmations.

## A12. Ambiguity volume

A quantitative ambiguity space belongs to one declared domain $d$ with state space $\mathcal{X}_d$ and explicit encoding.

At time $t$, define

$$\Omega_t^{(d)} = \{z \in \mathcal{X}_d : z \text{ remains compatible with } E_t \text{ and } C_t\},$$

where $E_t$ is available evidence and $C_t$ the active constraint set.

Let $\mu_d$ be a domain-specific measure and $\mu_{0,d}$ a reference measure. Define

$$A_{t,\text{vol}}^{(d)} = \log\left(1 + \frac{\mu_d(\Omega_t^{(d)})}{\mu_{0,d}}\right).$$

This quantity is dimensionless. States, paths, interpretations, identities, and hypotheses cannot be placed into one measured set unless a common encoding has been defined.

Let

$$R_t \in [0,1], \qquad X_t \in [0,1]$$

be normalised low-cost route connectivity and preserved access respectively. The working manoeuvrability index is

$$M_t = A_{t,\text{vol}}^{(d)} R_t X_t.$$

$M_t$ is a heuristic multiplicative audit index rather than a validated universal law. The product form should be compared with additive, interaction, and nonlinear alternatives.

See Ambiguity Dynamics and Manoeuvre Space.

# A13. Cross-layer address map

Let $K$ be a candidate invariant and $L \in \{S, I, P, O\}$ the active layer. Let $\theta_L$ denote layer-specific constraints, observables, units, noise terms, and admissible transitions.

$$K_L = A_L(K; \theta_L).$$

$A_L$ is an address map, not a new ontology.

A valid address map preserves the defining relation of $K$, explicit variable changes, layer-appropriate evidence, independent falsifiers, and a stated coupling where cross-layer influence is claimed.

Repeated algebraic form across layers does not establish the same physical mechanism. Formal cross-layer work therefore declares typed spaces and couplings, for example

$$C_{LM} : \mathcal{X}_L \to \mathcal{X}_M.$$

See Cross-Layer Invariants and Layer Addressing and Typed Traversal and Equation Hygiene.

# A14. Agency accessibility

Let $U_t$ be the actions actually available and $\hat{U}_t$ the actions perceived as available. Let

$$G_t(u; T_t, H_t) \in [0, 1], \qquad \theta_{\text{access}} \in [0, 1],$$

where $T_t$ represents perceived threat and $H_t$ reinforcement history.

The practically accessible set is

$$U_t^{\text{access}} = \{u \in U_t : G_t(u; T_t, H_t) > \theta_{\text{access}}\}.$$

Let $a_t \in [0, 1]$ represent practical accessibility:

$$\text{Agency}_{\text{effective}}(t) = \text{Agency}_{\text{capacity}}\, a_t.$$

This is an operational distinction rather than a moral, legal, or clinical equation.

See Agency Accessibility and Capture Geometry.

# A15. Path cost and weighting

For a trajectory $\gamma$ in one declared state space, a candidate path cost is

$$C[\gamma] = \int_\gamma \lambda(x)\, ds.$$

The path element $ds$, local cost $\lambda$, and resulting units must be declared. This integral cannot be applied across incompatible S, I, P, and O spaces without a typed common construction.

A Gibbs-like path weighting requires a dimensionless exponent. Using a dimensionless normalised cost $\tilde{C}[\gamma]$,

$$P(B \mid A) = \frac{1}{Z_A} \sum_{\gamma \in \Gamma(A \to B)} \exp\left[-\tilde{C}[\gamma]\right].$$

Alternatively, with inverse cost scale $\beta$,

$$P(B \mid A) = \frac{1}{Z_A} \sum_{\gamma \in \Gamma(A \to B)} \exp[-\beta C[\gamma]],$$

where

$$Z_A = \sum_{B'} \sum_{\gamma \in \Gamma(A \to B')} \exp[-\beta C[\gamma]].$$

Path multiplicity and cost are candidate predictors rather than a universal definition of probability.

## A16. Reduction rules

- If $R_t$ or $X_t$ is negligible, high ambiguity does not imply high manoeuvrability.
- If one layer address fails, evidence from another address cannot conceal the failure.
- If perceived and actual action sets do not differ, the agency-accessibility mechanism is unnecessary.
- If simpler noise, incentive, habit, uncertainty, or standard physical models perform better, they remain the preferred explanation.
- A mathematical expression is not evidence merely because it can be written.
- A tuple, projection, or named update map does not prove that its component spaces or couplings have been derived.

## A17. Related public documents

- MKUFT Core Extended
- Standalone Formal Addendum
- Experimental Test Programme
- Worked Examples: RNG and Environment
- Falsification Summary
- Typed Traversal and Equation Hygiene
- Cross-Support and Traversal Map

## Notation

The symbols $I$, $\mathcal{I}$, and $i$ are kept distinct for the information layer, information-function space, and one information structure respectively. Domain-qualified symbols are used where an unqualified symbol would hide incompatible spaces.